The Humanitarian Planner's Dilemma

Rapid, Evidence-Grounded Crisis Response Under Data Scarcity

Aaron Scherf Columbia University | Sustainable Development (SIPA) *3R Framework: Resilience, Response, and Recovery*

- **The Core Tension:** Extreme operational urgency vs. high evidence search costs and acute data scarcity.
- **The Goal:** An open-source, automated pipeline to ingest multi-modal data, quantify global evidence, and simulate intervention outcomes.



The Reality of Crisis Response: Search Costs & Portfolio Decisions

- **Evidence Search & Consumption Costs:**
  - Thousands of evaluations exist (3ie, J-PAL, World Bank), but they are locked in fragmented repositories and 80-page PDFs.
  - Planners cannot consume or standardize them in an emergency.
- **The Portfolio Optimization Problem:**
  - Academics isolate single interventions for internal validity.

*Responders have an **open canvas and fixed budget**—needing to find the* optimal mix* of cash, food aid, insurance, and logistics.

- **The Resilience Planning Parallel:**
  - Planners face severe data scarcity (no field surveys).
  - Almost always planning for **unprecedented local events**—inherently requiring evidence borrowed from other contexts.



State of the Art: The Limitations of Existing Frameworks

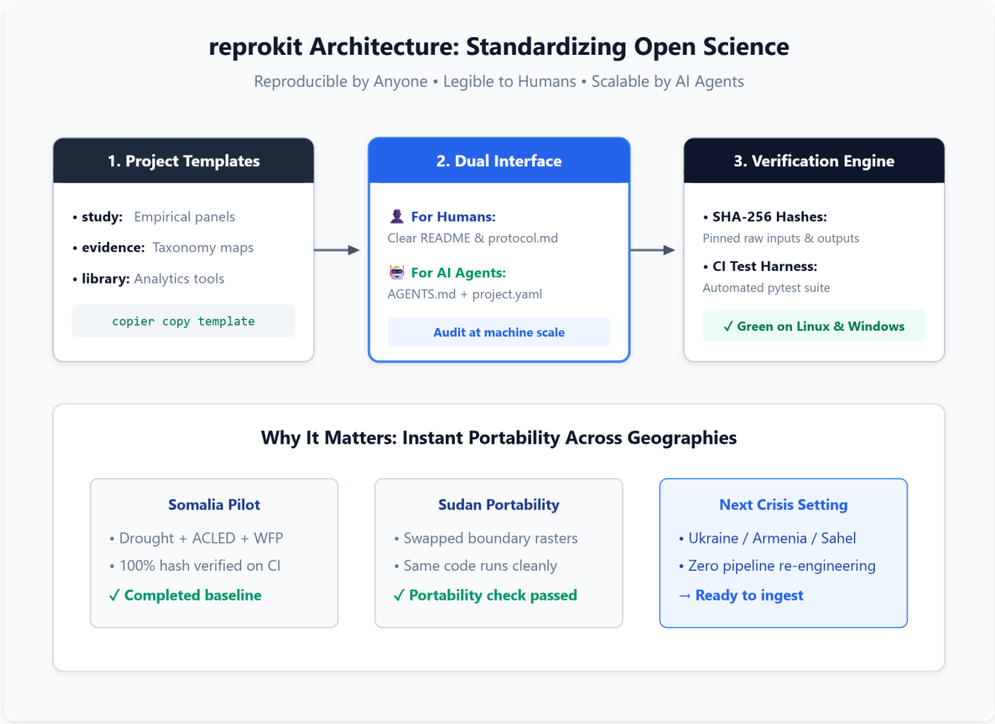
- **3ie Evidence & Gap Maps:**
  - Excellent taxonomies, but static and manually curated.
  - Disconnected from live spatial data; does not extract reusable estimators.
- **World Bank DIME / Microdata & HDX:**
  - Passive data repositories lacking counterfactual simulation.
- **Academic Hierarchical Modeling:**
  - Meager (2019) & Vivalt (2020): Prove high cross-study effect variance across microcredit and social programs.
  - Pritchett & Sandefur (2015): External trial estimates often yield higher error than local non-experimental data.

Limitation:* Retrospective academic studies on clean survey microdata; inapplicable to rapid, live crisis response.



Foundational Architecture: Standardized, Reproducible Research (`reprokit`)

- **Standardizing Open Science & Data Science:**
  - Orchestrated via `reprokit` (v0.4.0 released, open-source).
  - Independent verification regardless of technical background.
- **Three Project Archetypes:**
  - `study`: Empirical pipelines with automated hash verification.
  - `evidence`: Machine-readable evidence maps and manifests.
  - `library`: Reusable data fusion and analytics packages.
- **Built for Human and Agent Readability:**
  - Metadata files (`project.yaml`, `AGENTS.md`) allow autonomous LLM agents to parse and audit codebases at scale.
  - Dual-interface: accessible to humans, verifiable by agents.
- **Portability Across Contexts:** Hash-pinned inputs and CI on Linux/Windows enable rapid porting to new crises.



The 2×2 Evidence Engine: System Architecture

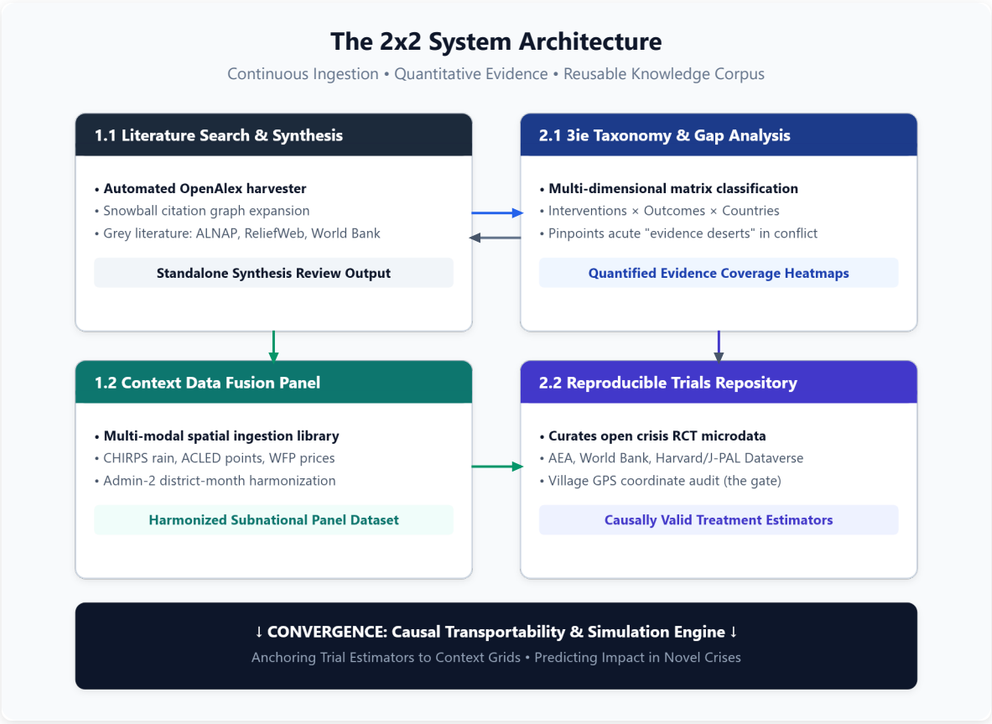
- **Dimension 1: Content Type**

Context Data:* Multi-modal physical, conflict, and market feeds. Evidence Data:* Published academic papers and institutional evaluations.

- **Dimension 2: Methodological Mode**

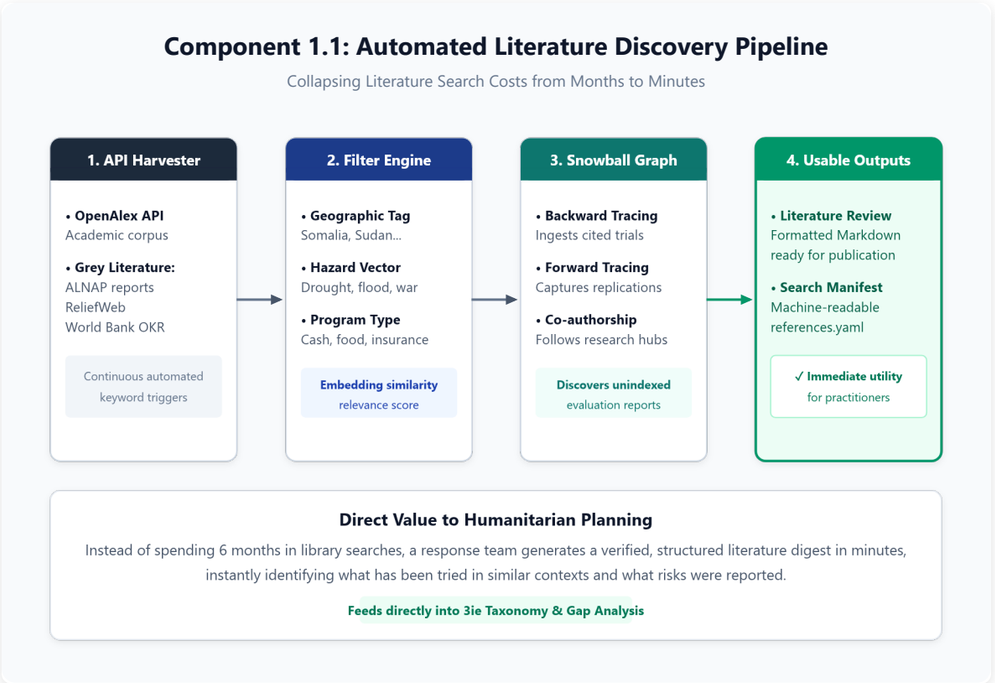
Discovery:* Continuous automated harvesting. Quantification:* Standardized taxonomies and microdata extraction.

- **The Four Quadrants:**
  - **1.1 Literature Search & Synthesis:** OpenAlex & web scraping.
  - **1.2 Context Data Fusion Panel:** Multi-modal subnational grids.
  - **2.1 3ie Evidence Taxonomy:** Quantifying coverage & gaps.
  - **2.2 Reproducible Trials Repository:** Open RCTs & estimators.



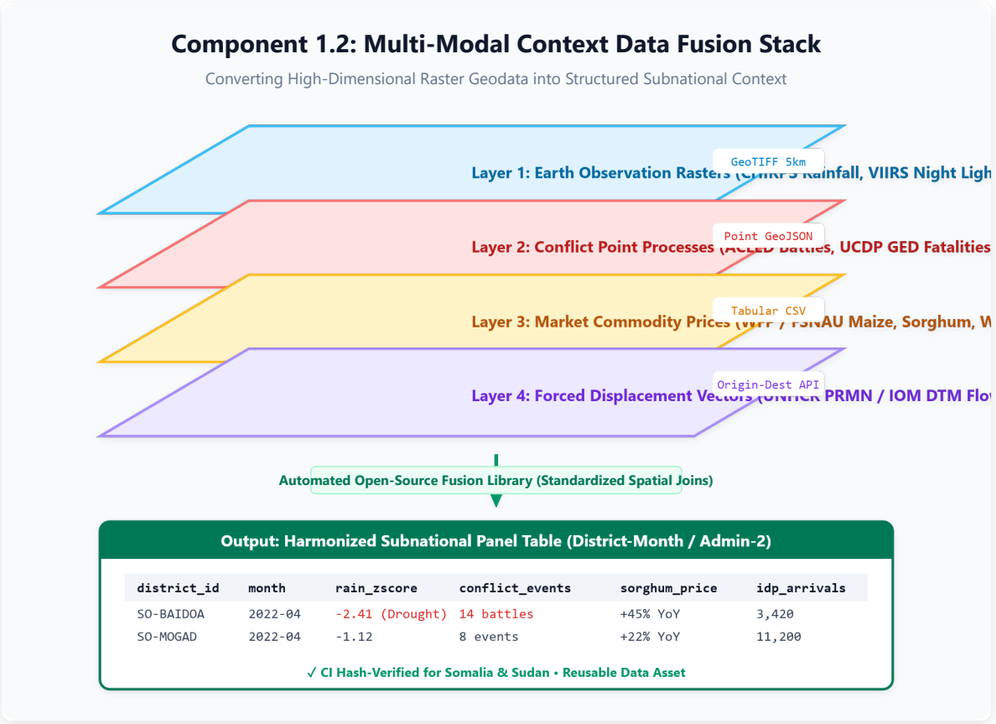
Component 1.1: Automated Literature Discovery & Synthesis

- **Overcoming Search Costs:**
  - Automated scrapers continuously harvest OpenAlex, academic APIs, and grey literature portals (ALNAP, ReliefWeb, World Bank).
- **Snowball & Semantic Graph Expansion:**
  - Traces forward/backward citation networks around key crisis topics.
  - Extracts metadata, abstract embeddings, and author linkages.
- **Standalone Usable Deliverables:**
  - Not just an intermediate step: produces a **comprehensive, publishable literature review** and evidence synthesis on demand for any designated geography.
- **Current Status:**
  - INITIALIZED Active harvesting; expanding from ad-hoc sampling to systematic corpus extraction.



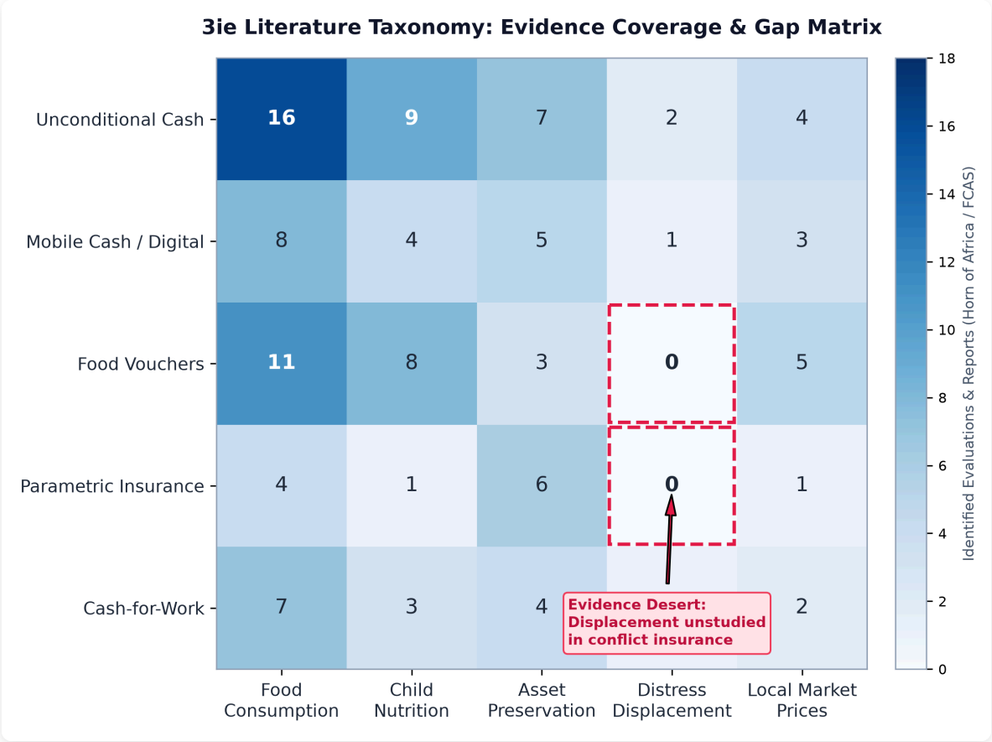
Component 1.2: Subnational Context Data Fusion & Analytics Library

- **Standalone Research Deliverables:**
  - The harmonized panel dataset and the data processing pipeline are standalone research assets.
- **The Open-Source Data Fusion Library:**
  - Automates multi-modal ingestion (satellite rasters, point events, price feeds).
  - Standardizes spatiotemporal boundaries (Admin-2 / district-month).
  - Automated cleaning, spatial matching, and missingness audits.
- **Core Technical Contribution:**
  - Transforms high-dimensional Earth observation rasters (CHIRPS precipitation, night lights) into structured, tabular context variables alongside conflict logs (ACLED/UCDP) and market feeds (WFP).
- **Proof of Concept:** Somalia and Sudan pipelines rebuilt with 100% hash matching on CI.



Component 2.1: Quantifying Evidence Coverage via 3ie Taxonomies

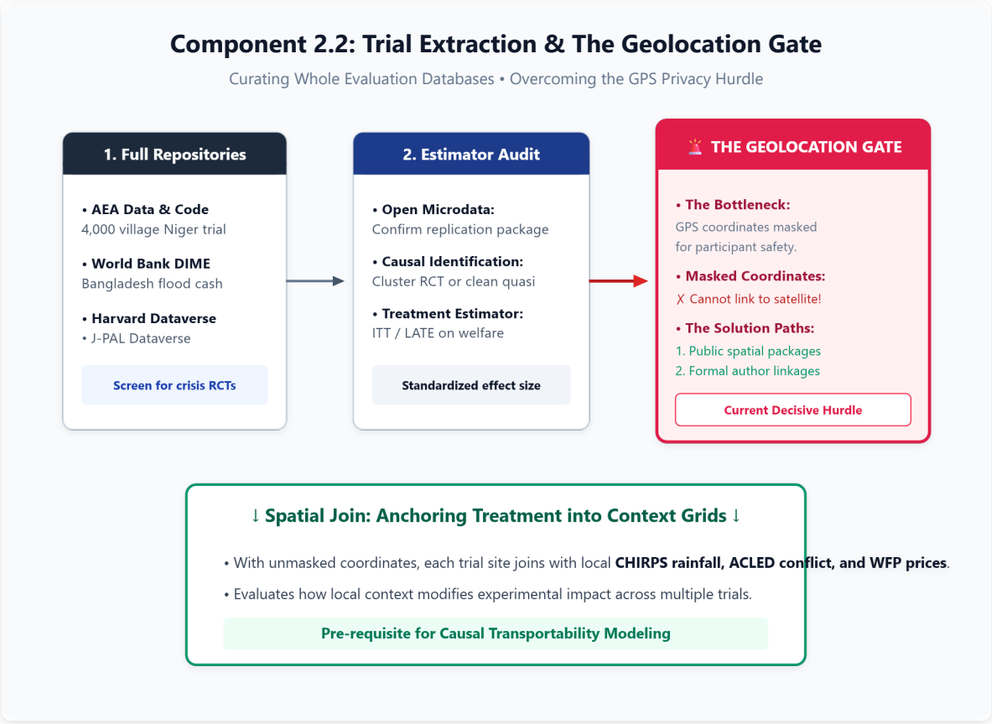
- **Multi-Dimensional Evidence Matrices:**
  - Categorizes academic and institutional grey literature using 3ie's standardized intervention (cash, vouchers, insurance, public works) and outcome taxonomies (consumption, nutrition, displacement).
- **Country-Specific Granularity:**
  - Generates distinct evidence heatmaps for any target country of interest (starting with Somalia and Sudan testbeds).
- **Diagnosing "Evidence Deserts":**
  - Dense green zones: Cash in peaceful agricultural settings.
  - Blank/red deserts: Parametric insurance, anticipatory cash, or social safety nets under active conflict.
- **Current Status:**
  - YELLOW Proof-of-concept classification complete on initial paper sample; scaling across the corpus.



Component 2.2: Curating Whole Trial Repositories & The Geolocation Gate

- **Mining Whole Evaluation Repositories:**
  - Drawing directly from full database corpuses:
  - AEA Data & Code (e.g., Niger cash/conflict rollout across 4,000 villages).
  - World Bank Microdata / DIME (e.g., Bangladesh flood cash).
  - Harvard Dataverse & J-PAL Dataverse (hundreds of open evaluation packages).
- **Extracting Estimators:**
  - Curating studies with open microdata and causally valid treatment estimators.
- **The Decisive Geolocation Gate:**
  - GPS coordinates are often masked in conflict zones for participant safety.

Core Challenge:* Obtaining unmasked village-level coordinates to link experimental treatment effects to satellite context grids.

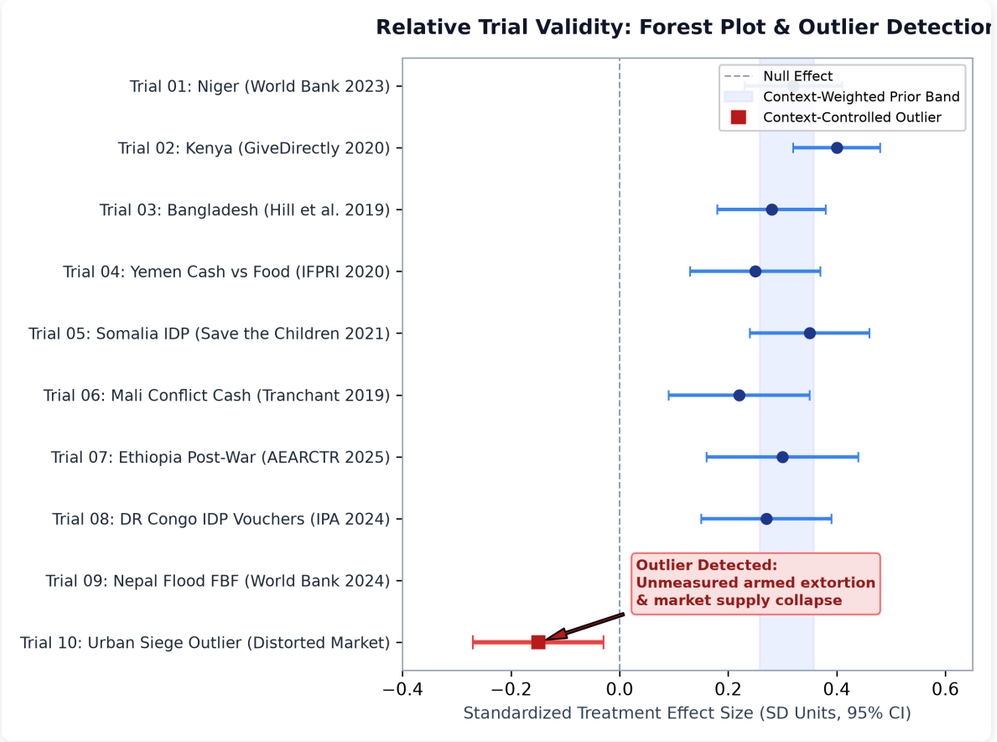


The Evidence Cycle: Iterative Feedback & Relative Trial Validity

- **Reinforcing Iterative Loops:**

Taxonomy $\leftrightarrow$ Search:* 2.1 gaps direct 1.1 targeted literature sweeps. Trials $\leftrightarrow$ Fusion:* 2.2 trials inject context variables into 1.2; 1.2 fusion panels anchor 2.2 trial coordinates.

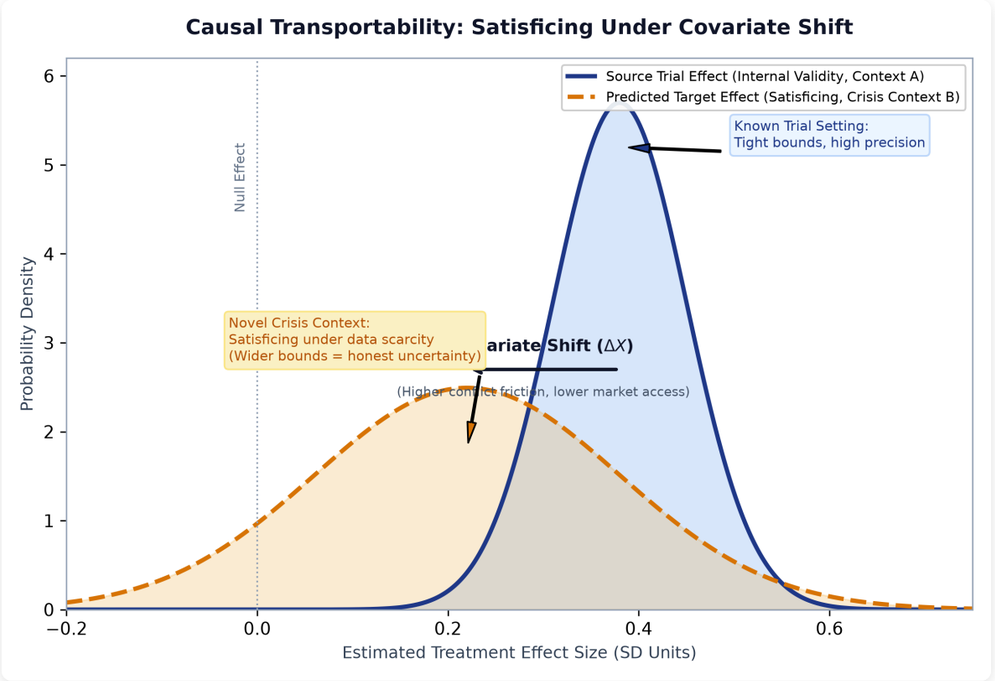
- **Asymptotic Convergence:**
  - Continuously expanding until evidence coverage for a target domain is as complete and searchable as possible.
- **Context-Aware Relative Validity Checks:**
  - Evaluating multiple trials of the same intervention across comparable context grids reveals relative validity.
  - If 9 trials show positive effects but 1 is an extreme outlier under the same context, it exposes a study flaw or unmeasured confounder.
- **Calibrating Priors:** Similarity weighting across related trials establishes empirical variance spreads for Bayesian priors.



Causal Transportability in Practice: Satisficing Under Covariate Shift

- **The External Validity Dilemma:**
  - Causal transportability is notoriously difficult even across peaceful settings (Pritchett & Sandefur 2015; Vivalt 2020).
  - In crises, where primary data collection is impossible, waiting for perfect evidence is fatal.
- **Satisficing Under Data Scarcity:**
  - Generating the best, most reliable impact estimates possible with empirically grounded confidence bounds.
- **The Standard Covariate Shift Problem:**
  - Holding the intervention functionally constant (e.g., standard cash transfer).
  - Context distributions shift (climate stress, market density, baseline poverty).
- **Measuring Non-Applicability:**

Estimating the degree to which existing evidence does not* apply in a new setting by modeling impact variability across known environments.

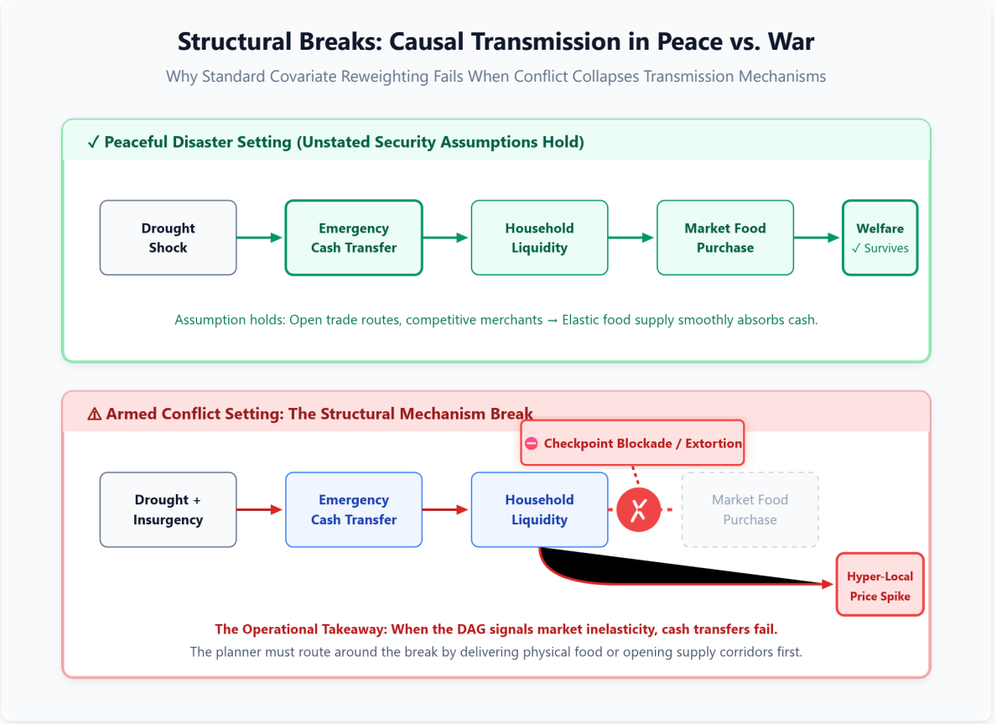


Beyond Covariate Shift: Structural Breaks in Crisis Environments

- **The Failure of Unstated Assumptions:**

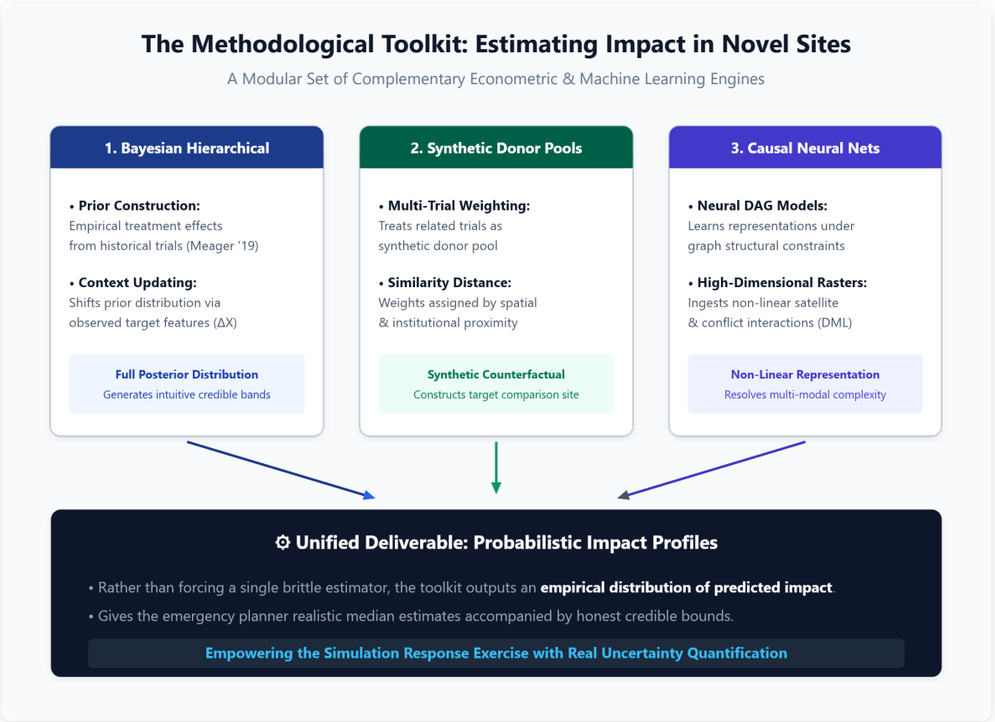
In Peace:* Cash to farmers increases drought resilience, assuming open trade corridors, functional mobile networks, and non-predatory authorities. In Conflict:* Armed groups extort recipients, mobile networks are jammed, or trade routes are blockaded. Causal linkages collapse.

- **Structural Break as a Validity Check:**
  - An essential audit of whether the causal chain itself is intact before attempting covariate adjustment.
- **Modeling Breakdowns with Directed Acyclic Graphs (DAGs):**
  - Mapping nodes: $Transfer \rightarrow Liquidity \rightarrow Market \; Purchase \rightarrow Welfare$.
  - Identifying friction nodes: Military blockades, inflation spikes, and extortion.
- **Operational Route-Arounds:** If the market link is broken (inelastic supply), planners know cash will fail and in-kind food distribution is required.



The Methodological Toolkit: Estimating Impact in Novel Localities

- **A Modular Set of Proposed Tools (Not a Single Estimator):**
- **Bayesian Hierarchical Modeling (e.g., Meager 2019):**
  - Priors from past trials, updated by observed context shifts (from fusion panel) to yield posterior distributions for the target site.
- **Synthetic Controls / Multi-Trial Weighting:**
  - Weighting historical trials as a synthetic donor pool based on spatial and institutional similarity.
- **Double Machine Learning (DML; Chernozhukov et al.):**
  - High-dimensional regularized estimation to control for complex spatiotemporal confounders.
- **Causal Neural Networks & Neural DAGs:**
  - Using causal deep learning (e.g., Dragonnet, Causal VAEs, Neural SEMs) to learn representations under DAG constraints, modeling non-linear interactions across satellite rasters.

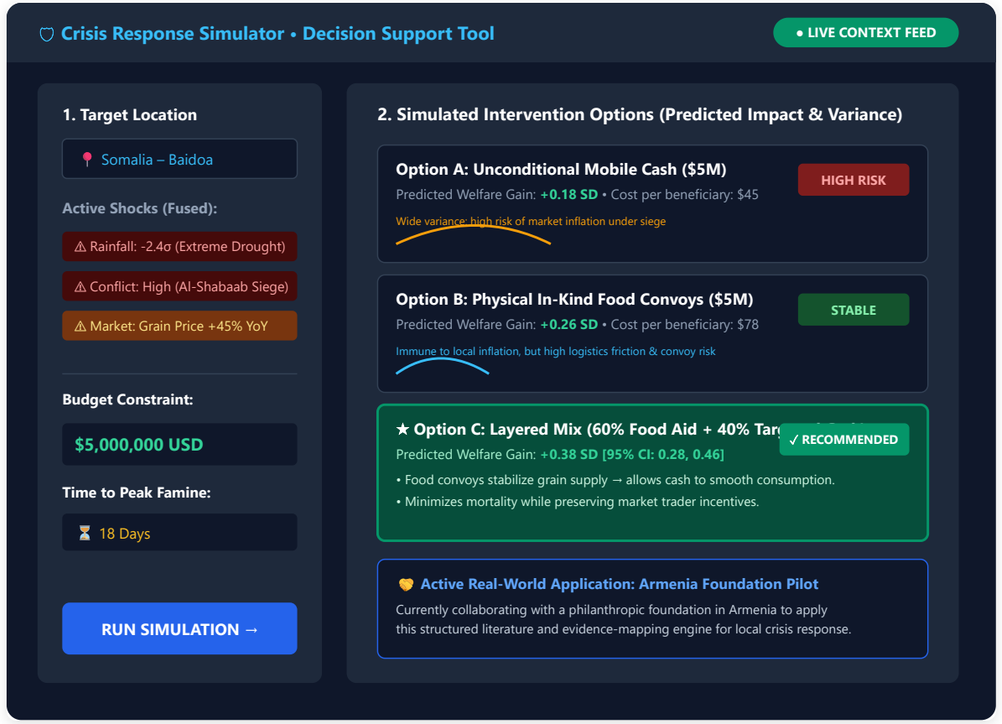


Decision Support in Action: The Simulation Response Exercise

- **The Simulation Response Exercise:**
  - Running the transport engine across multiple candidate interventions for the same crisis geography under a fixed budget:

Option A (Unconditional Cash):* High expected impact, but wide variance due to volatile local market prices. Option B (In-Kind Food Aid):* Lower logistical efficiency, but immune to market supply blockades. Option C (Parametric Index Payouts):* Rapid execution, but high basis risk.

- **Comparing & Layering Interventions:**
  - Modeling complementarities: evaluating whether layering cash with crop insurance or supply subsidies maximizes lives saved under budget caps.
- **Active Case Engagement:**
  - Ongoing practical collaboration with a foundation in Armenia to supply a structured evidence map and decision-support synthesis.



Methodological Frontiers, Open Questions & Roadmap

- **Core Open Questions:**

Intervention Combining & Layering:* How do we model complementarities across interventions in a novel context? Displacement & Migration Dynamics:* Does assistance facilitate safe evacuation or cause entrapment? Is displacement an outcome or a context filter? Spatial Resolution Limits:* Resolving conflict and displacement tracking below Admin-2 (district) to match village trial clusters. Proxy Surrogacy in Conflict:* When do satellite proxies (NDVI, night lights) fail statistical surrogacy for human welfare?

- **Ideal Testbeds:** Cash transfers and index insurance in drought-induced famine settings (Somalia and Sudan testbeds).
- **Immediate Roadmap:** Finalize village GPS linkages $\rightarrow$ Scale 3ie taxonomy classification $\rightarrow$ Complete transport proof of concept $\rightarrow$ Working paper for spring presentation.

